Here is ground-state electron momentum. Substitution gives

Ignoring electron interaction, the helium ground energy is times this value:

The ion energy minus atomic ground energy is the first ionization energy:

Method 2: the helium-ion energy is

Insert the ground angular momentum to rewrite :

Minimum ground energy makes the first term of zero. Thus radius and energy

are respectively as in .

Here are two methods for helium ground energy and radius .

Method 1: the mechanical equation

and ground-state quantization give the radius

Thus the helium ground energy is

Numerical substitution gives

Therefore the helium first ionization energy is

This still exceeds the measured helium ionization energy by